

2) Repeat ... Until :

- **problem 1 :** Write a treatment that reads a mark between 0 and 20, then displays it on 40.

Solution 1 : control of the mark with 'If... Else'

```
Print("Give a mark between 0 and 20"); Read(x);
```

```
If (x<0 or x>20) then
```

```
    Print ("Error : The mark is invalid");
```

```
Else
```

```
    x ← x*2;
```

```
    Print ("Your mark on 40 is : ", x);
```

```
EndIf;
```

Solution 2 : control of the mark with 'Repeat ... Until'

```
Repeat
```

```
    Print("Give a mark between 0 and 20"); Read(x);
```

```
Until (x>=0 and x<=20);
```

```
x ← x*2;
```

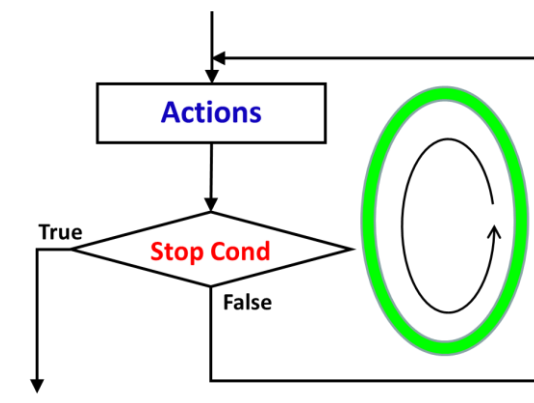
```
Print ("Your mark on 40 is : ", x);
```

Syntax :

```
Repeat
    Actions;
Until (stopping condition);
```

Qù :

stopping condition : It is a Boolean expression, which designates the case of stopping the repetition of actions.



- The number of repetitions is unknown, because it depends on a certain condition, we can Repeat the loop 1 time, 5 times, 1000000 times, for a week, a year,...etc. We leave the loop only when the condition is verified (is true).
- At the end of each iteration the condition is evaluated.
- The actions of the loop are executed at least once.

- **Problem 2 :** Write a treatment that calculates and displays the product of two real numbers as much as the user wants.

Solution :

Repeat

```
{1} Print("Give two real numbers : ");          Read (x, y);  
{2} res ← x*y; Print(x, " * ", y, " = ", res);  
{3} Print("Do you want to calculate the product of two other  
numbers if yes type 'O' else type 'N' :");  
      Read(Choice);
```

Until (Choice='N' or Choice='n');

Conclusion :

- ❖ We use the 'Repeat..Until' loop when the number of repetitions is at least equal to once.
- ❖ This loop is ideal when you want to control a data.

The data can be a value entered by the user (problem 1), or a choice expressed by the user (problem 2).